

Determine if the expressions are equivalent for the given values.

	Value	$7x + 3x - 6$	$2(5x - 8) + 10$	Equivalent?
1.	$x = -3$	$7(-3) + 3(-3) - 6$ $-21 + (-9) - 6$ $-30 - 6$ $= -36$	$2(5(-3) - 8) + 10$ $2(-15 - 8) + 10$ $2(-23) + 10$ $-46 + 10$ $= -36$	☒ yes ☐ no
2.	$x = 0$	$7(0) + 3(0) - 6$ $0 + 0 - 6$ $= -6$	$2(5(0) - 8) + 10$ $2(0 - 8) + 10$ $2(-8) + 10$ $-16 + 10$ $= -6$	☒ yes ☐ no
3.	$x = 0.3$	$7(0.3) + 3(0.3) - 6$ $2.1 + 0.9 - 6$ $3 - 6$ $= -3$	$2(5(0.3) - 8) + 10$ $2(1.5 - 8) + 10$ $2(-6.5) + 10$ $-13 + 10$ $= -3$	☒ yes ☐ no
4.	$x = \frac{1}{2}$	$7\left(\frac{1}{2}\right) + 3\left(\frac{1}{2}\right) - 6$ $\frac{7}{2} + \frac{3}{2} - 6$ $\frac{10}{2} - 6$ $5 - 6$ $= -1$	$2\left(5\left(\frac{1}{2}\right) - 8\right) + 10$ $2(2.5 - 8) + 10$ $2(-5.5) + 10$ $-11 + 10$ $= -1$	☒ yes ☐ no

5. Depending on the value being substituted into the expressions, the evaluated

expressions

☒ always
☐ sometimes
☐ never

had the same value as each other.

Determine whether the expressions are equivalent for the given values.

		$\frac{1}{3}(9 + 6y) + 4x$	$2\left(2x + y + \frac{3}{2}\right)$	Equivalent?
6.	$x = -2$ $y = 6$	$\frac{1}{3}(9 + 6 \cdot 6) + 4(-2)$ $\frac{1}{3}(9 + 36) - 8$ $\frac{1}{3}(45) - 8 \rightarrow \frac{45}{3} - 8$ $15 - 8 = 7$	$2\left(2(-2) + 6 + \frac{3}{2}\right)$ $2\left(-4 + 6 + \frac{3}{2}\right)$ $2\left(2 + \frac{3}{2}\right)$ $2(3.5) = 7$	☐ yes ☐ no
7.	$x = 0$ $y = 1$	$\frac{1}{3}(9 + 6 \cdot 1) + 4(0)$ $\frac{1}{3}(9 + 6) + 0$ $\frac{1}{3}\left(\frac{15}{1}\right)$ $= 5$	$2\left(2 \cdot 0 + 1 + \frac{3}{2}\right)$ $2\left(1 + \frac{3}{2}\right)$ $2(2.5)$ $= 5$	☐ yes ☐ no
8.	Answers will vary $x = 2$ $y = 0$	$\frac{1}{3}(9 + 6 \cdot 0) + 4(2)$ $\frac{1}{3}\left(\frac{9}{1}\right) + 8$ $3 + 8 = 11$	$2\left(2 \cdot 2 + 0 + \frac{3}{2}\right)$ $2\left(4 + \frac{3}{2}\right)$ $2(5.5) = 11$	☐ yes ☐ no

Determine if the following algebraic expressions are equivalent using any method you choose.

	Expression A	Expression B	Equivalent?
9.	$x(9 - 11) - \frac{3}{5}(10 + 15x)$ $-2x \quad -6 - 9$ $-2x - 15$	$-11x - 15$	☐ yes ☐ no
10.	$\frac{5}{9}(6x + 12x) - (25 + 3x)$ $\frac{5}{9}(18x) - 25 - 3x$ $10x - 25 - 3x$ $7x - 25$	$-25 + 10x$	☐ yes ☐ no